

# Phishing Email Analysis & Threat Investigation

This project demonstrates a SOC Analyst-style investigation of a phishing email. The analysis focuses on identifying malicious indicators, investigating suspicious links, reviewing email headers, and documenting Indicators of Compromise (IOCs).

## Project Objectives

- Identify phishing indicators
- Analyze suspicious email headers
- Detect spoofed domains and malicious links
- Investigate Indicators of Compromise (IOCs)
- Document mitigation recommendations

## Tools Used

| Tool       | Purpose                            |
|------------|------------------------------------|
| VirusTotal | Threat intelligence & URL scanning |
| MXToolbox  | Email header analysis              |
| Wireshark  | Packet analysis                    |
| URLScan.io | URL investigation                  |
| Any.Run    | Sandbox malware analysis           |

## Phishing Scenario

An employee received an email claiming to be from Microsoft requesting an urgent password reset. The message contained a suspicious login link and attempted to create urgency to steal credentials.

## Investigation Process

**Email Analysis:** Identified spoofed sender address, urgent language, and suspicious hyperlinks.

**Header Analysis:** SPF and DKIM validation failures were detected during email header review.

**URL Investigation:** The embedded URL redirected users to a fake Microsoft login page.

**Threat Intelligence:** VirusTotal flagged the domain as malicious and recently registered.

## Indicators of Compromise (IOCs)

| Type             | Indicator                   |
|------------------|-----------------------------|
| Malicious Domain | microsoft-login-security.ru |
| Suspicious IP    | 185.xxx.xxx.xxx             |

|                |                       |
|----------------|-----------------------|
| Attack Type    | Credential Harvesting |
| Email Spoofing | Detected              |

## Threat Assessment

**Severity:** High Risk

The phishing email attempted credential theft using a spoofed Microsoft login page. Multiple malicious indicators confirmed the email as a phishing attempt.

## Mitigation Recommendations

- Block malicious domains and IP addresses
- Quarantine the phishing email
- Reset affected user credentials
- Enable Multi-Factor Authentication (MFA)
- Conduct phishing awareness training

## Conclusion

This project demonstrates practical SOC Analyst skills including phishing detection, IOC analysis, email investigation, and incident reporting. The investigation process helped identify a high-risk phishing attempt and provided mitigation strategies to reduce organizational risk.